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Connector having a base including a partition wall between adjacent terminals which have non-plated portions partially disposed within the base

Abstract

A connector includes a base and a plurality of terminals, each of the terminals includes a plated portion in which a base material is covered with a film and a non-plated portion in which the base material is not covered with a film, the non-plated portion is disposed within the base, a portion of the non-plated portion is exposed from the surface of the base, and the base includes a partition wall that protrudes from the surface of the base between two adjacent terminals among the terminals.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2022-023203 filed on Feb. 17, 2022, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a connector with a plurality of terminals.

2. Description of Related Art

(3) Japanese Unexamined Patent Application Publication No. 2014-209836 (JP 2014-209836 A) discloses a power conversion device mounted on a vehicle. In the power conversion device, a plurality of terminals is disposed to be spaced apart from each other.

SUMMARY

(4) With the above technology, condensation may occur around the terminal depending on the state and environment of a vehicle. Each of the terminals is subjected to plating for forming a film to improve corrosion resistance. However, in the terminal manufacturing process, there are cases where a process is employed in which a non-plated portion where no film is disposed is formed on the surface of the terminal. In this case, when condensation adheres to the non-plated portions of two adjacent terminals, corrosion due to electrolytic corrosion may occur in the non-plated portions.

(5) The present disclosure provides a technique for suppressing the occurrence of electrolytic corrosion in a plurality of terminals having non-plated portions.

(6) A first aspect of the present disclosure relates to a connector including a base and a plurality of terminals. The terminals are held by the base and protruding from a surface of the base, the terminals being disposed to be spaced apart from each other. Each of the terminals includes a plated portion in which a base material is covered with a film and a non-plated portion in which the base material is not covered with a film and the non-plated portion is disposed within the base. A portion of the non-plated portion is exposed from the surface of the base. The base includes a partition wall that protrudes from the surface of the base between two adjacent terminals among the terminals.

(7) According to the first aspect of the present disclosure, it is possible to separate the non-plated portions of the two adjacent terminals by the partition wall. As a result, even when condensation adheres to the surface of the base, it is possible to restrain the condensation from adhering to the non-plated portions of the two adjacent terminals by the partition wall extending from the surface of the base. Accordingly, electrolytic corrosion in the non-plated portion can be suppressed.

(8) In the first aspect of the present disclosure, the height of the partition wall from the surface of the base may be equal to or greater than a distance between the two adjacent terminals.

(9) According to the first aspect of the present disclosure, in a situation where condensation adheres to the surface of the base, the condensation may be formed in a hemispherical shape. By setting the height of the partition wall to be equal to or greater than the distance between the adjacent terminals, it is possible to restrain the condensation from running over the partition wall and contacting both adjacent connectors.

(10) In the first aspect of the present disclosure, a tip of the partition wall may be formed with a curved surface.

(11) According to the first aspect of the present disclosure, it is possible to restrain condensation from accumulating on the upper end of the partition wall.

(12) The connector according to the first aspect of the present disclosure may further include a

resin covering portion that is disposed on the surface of the base and covers a portion of the non-plated portion exposed from the surface of the base.

(13) According to the first aspect of the present disclosure, it is possible to suppress contact of condensation with the non-plated portion. Accordingly, it is possible to suppress the occurrence of corrosion in the non-plated portion.

(14) In the first aspect of the present disclosure, in each of the terminals, the plated portion may get inclined downward at a portion that is further away from the base than the non-plated portion.

(15) In the first aspect of the present disclosure, the plated portions of the terminals may extend obliquely with respect to the base.

(16) A second aspect of the present disclosure relates to a connector including a base and a plurality of terminals held by the base and protruding from a surface of the base, the terminals being disposed to be spaced apart from each other. Each of the terminals includes a plated portion in which a base material is covered with a film and a non-plated portion in which the base material is not covered with a film. The non-plated portion is disposed to be spaced apart from the base.

(17) According to the second aspect of the present disclosure, it is possible to restrain condensation adhering to the surface of the base from coming into contact with the non-plated portion. As a result, it is possible to suppress occurrence of the situation where condensation adheres to the non-plated portions of two adjacent terminals. Accordingly, electrolytic corrosion in the non-plated portion can be suppressed.

(18) In the second aspect of the present disclosure, a distance from the surface of the base to the non-plated portion may be equal to or greater than a distance between the non-plated portions of the two adjacent terminals.

(19) According to the second aspect of the present disclosure, in a situation where condensation adheres to the surface of the base, the condensation may be formed in a hemispherical shape. By setting the distance from the surface of the base to the non-plated portion to be equal to or greater than the distance between the adjacent terminals, it is possible to suppress contact of the condensation adhering to the surface of the base with the non-plated portion.

(20) In the second aspect of the present disclosure, in each of the terminals, the plated portion may get inclined downward at a portion that is further away from the base than the non-plated portion.

(21) According to the second aspect of the present disclosure, it is possible to promote condensation adhering to the plated portion to run along the downwardly inclined plated portion to move to the opposite side of the non-plated portion.

(22) In the second aspect of the present disclosure, the plated portions of the terminals may extend obliquely with respect to the base.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

(2) FIG. 1 is a plan view of a connector of a first embodiment;

(3) FIG. 2 is an enlarged view of a region II of FIG. 1;

(4) FIG. 3 is an enlarged view of a region II of a connector of a second embodiment;

(5) FIG. 4 is an enlarged view of a region II of a connector of a third embodiment;

(6) FIG. 5 is a plan view of a connector of a fourth embodiment;

(7) FIG. 6 is an enlarged view of a region VI of FIG. 5; and

(8) FIG. 7 is an enlarged view of a region II of a connector of a fifth embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

(9) The technical elements of a connector that are aspects of the present disclosure are listed below. Each of the following technical elements is independently useful.

First Embodiment

(10) As shown in FIG. 1, a connector **10** of the present embodiment is used for electrical connection of devices mounted on a battery electric vehicle. For example, the connector **10** is a connector that is disposed in a charger (that is, an electric vehicle (EV) charger) for charging a battery mounted on the battery electric vehicle and is connected to an in-vehicle wire for supplying power from an external power supply. The connector **10** may be used for devices other than the battery electric vehicle.

(11) The connector **10** is disposed in a device (not shown). The connector **10** includes a cover **12**, a plurality of terminals **14**, and a base **20**. In the following, the term “terminals **14**” is used to collectively represent the “plurality of terminals **14**”, and the term “terminal **14**” is used to represent “any one of the plurality of terminals **14**”. The terminals **14** have the same configuration as each other. The terminals **14** are arranged side by side at regular intervals. The terminal **14** is made of a conductive metal, such as a copper alloy, for example. The terminal **14** extends linearly. The terminals **14** are connected to a board (not shown) in a state of being held by the base **20**. The base **20** is made of insulating resin. The terminals **14** are fixed by being press-fitted into the base **20**.

(12) The terminals **14** and the base **20** are covered with the cover **12**. Some of the tips of the terminals **14** are exposed from the cover **12**. The cover **12** is attached to the device with the connector **10**.

(13) FIG. 2 shows an enlarged view of three terminals **14a**, **14b**, **14c** around a surface **20a** of the base **20**, among the terminals **14**. In addition, hereinafter, when each terminal **14** of the terminals **14** is described, the terminal is represented as “terminal **14**”, and when specifying one of the terminals **14a**, **14b**, **14c**, is described, the terminal is represented as “terminal **14a**”, “terminal **14b**”, or “terminal **14c**”. The terminal **14** includes a plated portion **30** and a non-plated portion **32**. The non-plated portion **32** is disposed at an intermediate position in a longitudinal direction of the terminal **14**. The intermediate position is a position that does not include both ends in the longitudinal direction of the terminal **14** (that is, a right and left direction in FIG. 2).

(14) A plated portion **30** is disposed at each of both ends of the non-plated portion **32** in the longitudinal direction of the terminal **14**. The plated portion **30** extends linearly. The non-plated portion **32** has a wider shape than the plated portion **30**. The entire outer surface of the plated portion **30** is covered with a corrosion-resistant thin film. The thin film covering the plated portion **30** is made of a highly corrosion-resistant material, such as gold (Au) or nickel (Ni). The material of the plated portion **30** may be different across the non-plated portion **32**. For example, the portion of the plated portion **30** exposed from the surface **20a** of the base **20** may be covered with a gold (Au) thin film, whereas the portion buried in the base **20** may be covered with a nickel (Ni) thin film.

(15) On the other hand, the non-plated portion **32** is covered with a corrosion-resistant thin film like the plated portion **30**, except both end surfaces **32a** located in a lateral direction perpendicular to the longitudinal direction of the terminal **14** (that is, a vertical direction in FIG. 2). The end surfaces **32a** are not covered with a thin film, and the base material of the terminal **14** is exposed.

(16) Next, a method for manufacturing the terminal **14** will be described. First, a flat plate is cut out by pressing. The terminals **14** after pressing are arranged in a row in the lateral direction of the terminals **14** and are connected to the adjacent terminals **14** at the non-plated portions **32**. Next, plating is performed while in a state in which the terminals **14** are connected to each other. Then, the non-plated portion **32** between the terminals **14** adjacent to each other is cut in parallel with the longitudinal direction. Accordingly, the terminal **14** is produced. In the manufacturing method, the non-plated portion **32** is cut after plating, and accordingly, the end surfaces **32a** of the non-plated portion **32** are not plated and the material of the terminal **14** is exposed.

(17) As shown in FIG. 2, a portion of the end surface **32a** is exposed from the surface **20a** of the base **20**.

(18) A partition wall **22a** extending along the terminals **14a**, **14b** protrudes from the surface **20a** of the base **20** between the terminals **14a**, **14b** adjacent to each other. The partition wall **22a** is disposed in the center of a gap between the terminals **14a**, **14b**. The partition wall **22a** is produced by integral molding with the base **20**. The partition wall **22a** extends linearly in a specific direction perpendicular to the longitudinal direction and the lateral direction of the terminal **14a** (that is, direction perpendicular to the paper surface of FIG. 2) with the shape being maintained in the specific direction. The partition wall **22a** extends over a section in which the terminals **14a**, **14b** are disposed in a specific direction.

(19) The width of the partition wall **22a** in the lateral direction of the terminal **14** gradually narrows as the distance in the longitudinal direction thereof becomes further away from the base **20**. A tip of the partition wall **22a** is formed with a curved surface and does not have a flat portion. A height **H** of the partition wall **22a** in the longitudinal direction is equal to or greater than an interval **L** between the terminals **14a**, **14b**. In addition, the height **H** may be, for example, 1.0 mm or more.

(20) A partition wall **22b** similar to the partition wall **22a** is disposed between the terminal **14b** and the terminal **14c**. As shown in FIG. 1, a partition wall **22** similar to the partition wall **22a** is disposed between two terminals **14** adjacent to each other among the terminals **14**. That is, the base **20** is provided with a plurality of partition walls **22** protruding from the surface **20a**.

(21) Condensation adheres to the connector **10** depending on the usage situation the surrounding environment. For example, moisture absorbed in a board to which the terminals **14** are connected is discharged from the board as the temperature of the board rises and turns into condensation. When condensation occurs, the condensation may adhere to the surface **20a** of the base **20**. When the condensation adheres to the surface **20a**, the end surface **32a** of the non-plated portion **32** is exposed to the condensation. The partition wall **22** is disposed in the connector **10**. With the above configuration, it is possible to restrain the end surfaces **32a** of the terminals **14** adjacent to each other (for example, the terminals **14a**, **14b**) from being electrically connected through moisture of the condensation. Accordingly, it is possible to restrain the terminal **14** from being corroded by electrolytic corrosion.

(22) The height **H** of the partition wall **22** is greater than the interval **L** between the terminals **14** adjacent to each other. With the above configuration, it is possible to suppress the water droplets of condensation adhering to the surface **20a** from crossing the partition wall **22**. For example, when the water droplets adhering to the surface **20a** are formed in a hemispherical shape, at the maximum, the water droplet has a hemispherical shape with a diameter equal to the distance between the partition walls **22** adjacent to each other, that is, the interval **L**, and the height of the water droplet is $L/2$. Therefore, by setting the partition wall **22** to at least twice the maximum height of the water droplet, it is possible to reliably restrain the water droplet from running over the partition wall **22**.

(23) The tip of the partition wall **22** is formed with a curved surface. With the above configuration, it is possible to restrain moisture due to condensation from accumulating at the tip of the partition wall **22**.

(24) The partition walls **22** are separated from each other and are not connected except for the base **20**. Accordingly, it is possible to restrain the moisture accumulated between the adjacent partition walls **22** from remaining between the adjacent partition walls **22**.

Second Embodiment

(25) As shown in FIG. 3, in a connector of the present embodiment, a covering portion **50** is disposed on the surface **20a** of the base **20** to cover the non-plated portion **32** of each of the terminals **14**. Other configurations of the connector **10** are the same as those of the connector **10** of the first embodiment. The covering portion **50** is made of a resin material, such as polybutylene terephthalate (that is, PBT). For example, the covering portion **50** may be produced by potting the

non-plated portions **32** of the terminals **14** from the state shown in FIG. 2.

(26) With the above configuration, effects similar to the effects of the connector **10** of the first embodiment can be obtained. Further, with the configuration, the covering portion **50** can restrain the non-plated portion **32** from being exposed to the outside of the connector **10**. Accordingly, corrosion of the non-plated portion **32** is suppressed.

(27) In a modification example, for example, of two adjacent terminals **14**, a first terminal **14** may be provided with the covering portion **50** and a second terminal **14** may not be provided with the covering portion **50** such that the terminal **14a** is provided with the covering portion **50**, the terminal **14b** is not provided with the covering portion **50**, and the terminal **14c** is provided with the covering portion **50**. In the configuration, the terminals **14** on which the covering portions **50** are disposed and the terminals **14** on which the covering portions **50** are not disposed may be alternately disposed in the direction in which the terminals **14** are disposed. With the configuration, it is possible to restrain the non-plated portions **32** of two adjacent terminals **14** from being electrically connected to each other due to moisture of condensation.

Third Embodiment

(28) As shown in FIG. 4, in a connector of the present embodiment, a plurality of terminals **114** are disposed in place of the terminals **14** as compared with the connector of the second embodiment. The terminal **114** includes a non-plated portion **32**, which is similar to the non-plated portion of the terminal **14**, and a plated portion **130** extending from each of both ends of the non-plated portion **32** in a longitudinal direction of the terminal **114**. The plated portion **130** extends obliquely with respect to the base **20**. In FIG. 4, the portion of the plated portion **130** that is further away from the base **20** than the non-plated portion **32** gets inclined downward as the distance from the non-plated portion **32** increases. All terminals **114** disposed on the connector, including terminals **114a**, **114b**, **114c**, have similar configurations to each other. Other configurations are the same as the configurations of the connector **10** of the second embodiment.

(29) With the above configuration, effects similar to the effects of the connector of the second embodiment can be obtained. Further, with the configuration, it is possible to promote the movement of a water droplet C of condensation adhering to the plated portion **130** in a direction away from the non-plated portion **32**. Accordingly, it is possible to restrain moisture from adhering to the periphery of the non-plated portion **32**.

Fourth Embodiment

(30) Differences from the first embodiment will be mainly described. As shown in FIG. 5, a connector **410** of the present embodiment includes a cover **412**, a plurality of terminals **414**, and a base **420**. The cover **412** has the same configuration as cover **12**.

(31) The terminals **414** have the same configuration as each other. The terminals **414** are disposed side by side at regular intervals. Similar to the terminal **14**, the terminal **414** includes a plated portion **430** similar to plated portion **30** and a non-plated portion **432** similar to non-plated portion **32**. However, the position of the non-plated portion **432** with respect to the plated portion **430** in a longitudinal direction of the terminal **414** is different from the position of the non-plated portion **32** with respect to the plated portion **30** in the longitudinal direction of the terminal **14**. The base **420** differs from the base **20** in that the partition walls **22** are not disposed, and other configurations of the base **420** are the same as configuration of the base **20**.

(32) The terminals **414** are connected to a board (not shown) in a state of being held by the base **420**. The terminals **414** are fixed by being press-fitted into the base **420**.

(33) FIG. 6 shows an enlarged view of three terminals **414a**, **414b**, **414c** around a surface **420a** of the base **420**, among the terminals **414**. In addition, hereinafter, when each terminal **414** of the terminals **414** is described, the terminal is represented as “terminal **414**”, and when specifying one of the terminals **414a**, **414b**, **414c**, is described, the terminal is represented as “terminal **414a**”, “terminal **414b**”, or “terminal **414c**”.

(34) The non-plated portion **432** is disposed at a position separated by a distance D from the

surface **420a** of the base **420**. The non-plated portion **432** is entirely exposed from the base **420**. Therefore, both end surfaces **432a** of the non-plated portion **432** are completely exposed from the base **420**. The distance D is greater than an interval L.

(35) With the above configuration, even when water droplets due to condensation adhere to the surface **420a** of the base **420**, it is possible to restrain the water droplets from reaching the non-plated portion **432**.

(36) Further, even when a water droplet C of condensation adhering to the surface **420a** has a hemispherical shape with a diameter of the interval L, the distance D from the surface **420a** to the non-plated portion **432** is two times or more the maximum height of the water droplet C.

Accordingly, it is possible to restrain water droplets from contacting the non-plated portion **432**.
Fifth Embodiment

(37) As shown in FIG. 7, in a connector of the present embodiment, a plurality of terminals **514** is disposed in place of the terminals **414** as compared with the connector of the fourth embodiment. The terminal **514** includes a non-plated portion **432**, which is similar to the non-plated portion of the terminal **414**, and a plated portion **530** extending from each of both ends of the non-plated portion **432** in a longitudinal direction of the terminal **514**. The plated portion **530** extends obliquely with respect to the base **420**. In FIG. 7, the portion of the plated portion **530** that is further away from the base **420** than the non-plated portion **432** gets inclined downward as the distance from the non-plated portion **432** increases. The same goes for all terminals **514** disposed in the connector, including terminals **514a**, **514b**, **514c**. Other configurations are the same as the configurations of the connector of the fourth embodiment.

(38) With the above configuration, effects similar to the effects of the connector of the fourth embodiment can be obtained. Further, with the configuration, as with the connector of the third embodiment, it is possible to promote the movement of the moisture of condensation adhering to the plated portion **530** in a direction away from the non-plated portion **432**. Accordingly, it is possible to restrain moisture from adhering to the periphery of the non-plated portion **432**.

(39) Although specific examples of the present disclosure have been described in detail above, things described above are merely examples and do not limit the scope of claims. The techniques described in the claims include various modifications and changes of the specific examples illustrated above.

(40) For example, in the fourth and fifth embodiments, two adjacent terminals (e.g., the terminal **414a** and the terminal **414b**) may have different distances from surface **420a** of the base **420**. With the above configuration, it is possible to increase the distance between the non-plated portions of two adjacent terminals. For example, the terminal **414a** may have a portion of the non-plated portion **432** buried in the base **420**, similar to the terminal **14a**.

(41) In the first to third embodiments, the partition wall **22** of the base **20** may not be disposed.

(42) In the third embodiment, the covering portion **50** may not be disposed.

(43) The technical elements described in the present disclosure or in the drawings exhibit their technical usefulness alone or in various combinations, and are not limited to the combinations described in the claims at the time of filing. In addition, the techniques exemplified in the present disclosure or the drawings can achieve a plurality of purposes at the same time, and achieving one of the purposes itself has technical usefulness.

Claims

1. A connector comprising: a base; and a plurality of terminals held by the base and protruding from a surface of the base, the terminals being disposed to be spaced apart from each other, wherein: each of the terminals includes a plated portion in which a base material is covered with a film, and a non-plated portion in which the base material is not covered with a film; the non-plated portion is disposed within the base; a portion of the non-plated portion is exposed from the surface

of the base; and the base includes a partition wall that protrudes from the surface of the base between two adjacent terminals among the terminals.

2. The connector according to claim 1, wherein a height of the partition wall from the surface of the base is equal to or greater than a distance between the two adjacent terminals.

3. The connector according to claim 1, wherein a width of the partition wall gradually narrows as the partition wall extends in a longitudinal direction thereof further away from the base, and a tip of the partition wall is formed with a curved surface.

4. The connector according to claim 1, further comprising a resin covering portion that is disposed on the surface of the base and covers a portion of the non-plated portion exposed from the surface of the base.

5. The connector according to claim 1, wherein in each of the terminals, the plated portion gets inclined downward at a portion that is further away from the base than the non-plated portion.

6. The connector according to claim 1, wherein the plated portions of the terminals extend, along a longitudinal direction thereof, obliquely with respect to the surface of the base from which the terminals protrude.

7. The connector according to claim 1, wherein the film is made of a metal.

8. The connector according to claim 7, wherein the metal is selected from the group consisting of gold (Au) and nickel (Ni).

9. The connector according to claim 7, wherein the plated portions of the terminals extend obliquely with respect to the surface of the base from which the terminals protrude, and the partition wall extends from the same surface of the base, in a same direction as and obliquely with respect to the two adjacent terminals between which the partition wall is arranged.

10. The connector according to claim 7, wherein, in each of the terminals, along a longitudinal direction of the terminal, the non-plated portion has a first portion, and a second portion, and the terminal has a first end, and a second end opposite to the first end, in a lateral direction perpendicular to the longitudinal direction of the terminal, both the first portion and the second portion of the non-plated portion are wider than the plated portion, the first portion of the non-plated portion and the first end of the terminal are buried in the base, and the second portion of the non-plated portion and the second end of the terminal are outside the base.

11. The connector according to claim 10, wherein, in each of the terminals, the base material is exposed at end surfaces of the second portion of the non-plated portion outside the base.

12. The connector according to claim 10, wherein, for a terminal among the terminals, the connector further comprises a resin covering portion that is disposed on the surface of the base, from which the terminal protrudes, and covers the second portion of the non-plated portion outside the base.

13. The connector according to claim 10, wherein, in each of the terminals, the plated portion extends from the second portion of the non-plated portion, obliquely with respect to both the second portion of the non-plated portion and the surface of the base, from which the terminal protrudes.

14. The connector according to claim 13, wherein a width of the partition wall gradually narrows as the partition wall extends in a longitudinal direction thereof further away from the base, and a tip of the partition wall is formed with a curved surface.

15. A connector comprising: a base; and a plurality of terminals held by the base and protruding from a surface of the base, the terminals being disposed to be spaced apart from each other, wherein: each of the terminals includes a plated portion in which a base material is covered with a film, and a non-plated portion in which the base material is not covered with a film; and the non-plated portion is disposed to be spaced apart from the base.

16. The connector according to claim 15, wherein a distance from the surface of the base to the non-plated portion is equal to or greater than a distance between the non-plated portions of the two adjacent terminals.

17. The connector according to claim 15, wherein in each of the terminals, the plated portion gets inclined downward at a portion that is further away from the base than the non-plated portion.
18. The connector according to claim 15, wherein the plated portions of the terminals extend, along a longitudinal direction thereof, obliquely with respect to the surface of the base from which the terminals protrude.
19. The connector according to claim 15, wherein the film is made of a metal.
20. The connector according to claim 15, wherein the metal is selected from the group consisting of gold (Au) and nickel (Ni).
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